

## **ENGR. DR. MUDASSAR AZAM**

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### **Academic Qualification**

**Ph.D. Chemical Engineering      2017-2020      (with distinction)**

Technical University of Vienna, Austria (TU WIEN)

**M.Sc. Chemical Engineering      2009-2011      (2<sup>nd</sup> position)**

Institute of Chemical Engineering & Technology, University of The Punjab, Lahore.

**B.Sc. Chemical Engineering 2003-2007      (5<sup>th</sup> position)**

Institute of Chemical Engineering & Technology, University of The Punjab, Lahore.

### **Teaching Experience:**

As a chemical engineering educator with more than **15 years** of experience, I aim to create an engaging and supportive environment where students can think critically, solve problems, and connect classroom learning to real-world challenges. I focus on making complex concepts easy to understand and encourage curiosity, adaptability, and teamwork. My goal is to equip students with the skills and confidence to succeed in their careers and make a positive impact on society.

**Assistant Professor**

**Jan 2012 to date**

**Institute of Chemical Engineering & Technology,  
University of the Punjab.**

Teaching (Undergraduate level) following subjects

- Instrumentation and Process Control
- Biofuel and Biorefinery
- Fluid and Particle Mechanics
- Separation Processes
- Heat Transfer
- Chemical Engineering Principles
- BioChemical Engineering Processes
- Thermodynamics

Teaching (Graduate level) following subjects

- Advanced Instrumentation and Process Control
- Advanced Separation Processes

**Lecturer**

**June, 2008 to Dec, 2011**

**Institute of Chemical Engineering & Technology,  
University of the Punjab.**

Same teaching assignments as mentioned above excluding graduate level.

### **Research Experience:**

My research in chemical engineering focuses on innovative approaches to waste management, energy conversion, and environmental sustainability. I have extensively

studied the characterization, combustion, and co-combustion of municipal solid waste (MSW) and refuse-derived fuels (RDF) for waste-to-energy applications, addressing critical challenges related to energy efficiency, circular economy, and environmental impact. Additionally, I have contributed to the development of advanced materials for energy storage and wastewater treatment, leveraging techniques such as kinetic modeling, catalytic processes, and material characterization to deliver sustainable solutions. My work aims to integrate multidisciplinary methods to address global challenges in renewable energy, pollution control, and resource recovery.

#### **Research Interests:**

- Development and optimization of waste-to-energy systems, including the utilization of MSW and RDF for sustainable energy generation.
- Investigation of advanced combustion and co-combustion techniques for solid waste and low-rank coal to improve energy efficiency and reduce emissions.
- Study of the synthesis and applications of nanomaterials, such as MXenes and metal-organic frameworks, for energy storage, water treatment, and environmental applications.
- Exploration of microbial fuel cells for simultaneous wastewater treatment and energy harvesting.
- Application of advanced material characterization methods to assess thermal, kinetic, and structural properties of energy and environmental systems.
- Development of innovative separation technologies and antifouling membranes for water treatment and desalination.

#### **Research Supervision:**

- 50 x B.Sc Students (Graduated)
- 08 x M.Sc Students (Graduated)
- 02 x PhD Students (Under Process)

#### **Research Grants:**

| PROJECT                                                                                                                                                       | FUNDING AGENCY    | GRANT          | ROLE   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------|--------|
| <ul style="list-style-type: none"> <li>• Biodegradation of Nitro-aromatic energetic materials by Using Bio Electrochemical Treatment System (BETS)</li> </ul> | Punjab University | 0.2 Million PK | PI     |
| <ul style="list-style-type: none"> <li>• Removal of Chromium from Industrial Waste Water using Microbial Fuel Cell</li> </ul>                                 | Punjab University | 0.3 Million PK | Co- PI |

#### **Administration:**

**Role:** Focal Person for Career Counselling & Placements

Key Responsibilities:

- Facilitating career development sessions and workshops to guide students on resume building, interview preparation, and job market trends.
- Establishing and maintaining relationships with industry partners to create internship and job placement opportunities for students.
- Organizing recruitment drives, career fairs, and on-campus placement events.

**Role:** Head Alumni Association ICET, PU.

Key Responsibilities:

- Building and maintaining strong relationships with alumni to foster a vibrant and engaged alumni community.
- Organizing events, reunions, and networking opportunities to connect alumni with the institution and current students.
- Establishing mentorship programs where alumni guide students in their academic

and professional journeys.

**Role:** Sports In charge

Key Responsibilities:

- Planning, organizing, and supervising sports events, tournaments, and recreational activities for students and staff.
- Coordinating with sports committees, coaches, and student teams to promote active participation in various sports.

**Other Roles:**

- Member of Green Youth Project by Prime Minister, PK,
- Member of Examination Cell
- Member of Outcome- Based Education System Team. ( ICET, PU.)
- Member of Doctoral Program Committee, (ICET,PU)
- Member of Continuing Professional Development Cell
- Head of Safety Training Program

### Industrial Experience:

**Fauji Fertilizer Company Limited, Pakistan**

**Sep 2007 to May 2008**

Worked as trainee Engineer in Fauji Fertilizer Company, GOTH MACHHI at Ammonia plant.

- Worked on all unit operations and unit processes of ammoniaplant.
- Worked on boiler and cooling tower.
- Training on DCS of Ammonia plant.
- Participated in Safety Training Program and work permits.
- Visited Urea section.
- Studied Laboratory Testing Procedures.

### Publications Index

|                     |        |
|---------------------|--------|
| Total impact Factor | = 175  |
| Citation            | = 1301 |
| h-index             | = 18   |
| i 10-index          | = 25   |

### Publications

1. **Azam, M.;** Jahromy, S.S.; Raza, W.; Raza, N.; Lee, S.S.; Kim, K.-H.; Winter, F. Status, characterization, and potential utilization of municipal solid waste as renewable energy source: Lahore case study in Pakistan. *Environment international* **2020**, *134*, 105291
2. **Azam, M.;** Ashraf, A.; Jahromy, S.S.; Raza, W.; Khalid, H.; Raza, N.; Winter, F. Isoconversional nonisothermal kinetic analysis of municipal solid waste, refuse-derived fuel, and coal. *Energy Science & Engineering* **2020**.
3. **Azam, M.;** Setoodeh Jahromy, S.; Raza, W.; Wesenauer, F.; Schwendtner, K.; Winter, F. Comparison of the Characteristics of Fly Ash Generated from Bio and Municipal Waste: Fluidized Bed Incinerators. *Materials* **2019**, *12*, 2664.
4. **Azam, M.;** Jahromy, S.S.; Raza, W.; Jordan, C.; Harasek, M.; Winter, F. Comparison of the combustion characteristics and kinetic study of coal, municipal solid waste, and refuse-derived fuel: Model-fitting methods. *Energy Science & Engineering* **2019**.
5. **Azam, M.** (2020). *Combustion of municipal solid waste (MSW) and refuse-derived fuel*

(RDF) with low rank coal; For waste to energy concepts. Technische Universität Wien,

6. **Azam, M.**, Ashraf, A., Setoodeh Jahromy, S., Miran, S., Raza, N., Wesenauer, F., Jordan, C., Harasek, M., & Winter, F. (2021). Co-combustion studies of low-rank coal and refuse-derived fuel: performance and reaction kinetics. *Energies*, 14(13), 3796.
7. Gul, H., Raza, W., Lee, J., **Azam, M.**, Ashraf, M., & Kim, K.-H. (2021). Progress in microbial fuel cell technology for wastewater treatment and energy harvesting. *Chemosphere*, 281, 130828.
8. Jordan, C., Setoodeh Jahromy, S., **Azam, M.**, Harasek, M., Winter, F., Vogtenhuber, H., Birkelbach, F., & Werner, A. (2020). Waste2Storage-Potenzial der Flugasche als thermochemischer Energie-& CO<sub>2</sub> Speicher.
9. Rizvi, S. F. J., Miran, S., **Azam, M.**, Arif, W., Wasif, M., & Garcia, H. P. (2021). Numerical Analysis of a Liquid Nitrogen (LN<sub>2</sub>) Engine for Efficient Energy Conversion. *ACS omega*, 6(24), 15663-15673.
10. Shahzad, A., Oh, J.-M., **Azam, M.**, Iqbal, J., Hussain, S., Miran, W., & Rasool, K. (2021). Advances in the synthesis and application of anti-fouling membranes using two-dimensional nanomaterials. *Membranes*, 11(8), 605.
11. Wesenauer, F., Jordan, C., **Azam, M.**, Harasek, M., & Winter, F. (2021). Considerations on Temperature Dependent Effective Diffusion and Permeability of Natural Clays. *Materials*, 14(17), 4942.
12. Raza, N.; Raza, W.; Gul, H.; **Azam, M.**; Lee, J.; Vikrant, K.; Kim, K.-H. Solar-light-active silver phosphate/titanium dioxide/silica heterostructures for photocatalytic removal of organic dye. *Journal of Cleaner Production* **2020**, 254, 120031.
13. Abbas, W., Abbas, S., Nawaz, M., **Azam, M.**, Oh, J.-M., & Shahzad, A. (2021). Development of polystyrene coated persulfate slow-release beads for the oxidation of targeted PAHs: effects of sulfate and chloride ions. *Journal of Hazardous Materials*, 416, 125879.
14. Ashraf, M., Ramzan, N., **Azam, M.**, Anwar, A., Khan, R. U., Durrani, A. K., & Rashid, M. U. (2024). Cattle dung conversion to syngas: solar photovoltaic integrated gasification system. *Biomass Conversion and Biorefinery*, 14(8), 9097-9108.
15. Raza, N., Raza, W., Gul, H., **Azam, M.**, Lee, J., Vikrant, K., & Kim, K.-H. (2021). Corrigendum for “Solar-light-active silver phosphate/titanium dioxide/silica heterostructures for photocatalytic removal of organic dye”[J. Clean. Prod. 254 (2020) 120031]. *Journal of Cleaner Production*, 281.
16. Raza, W.; Kukkar, D.; Saulat, H.; Raza, N.; **Azam, M.**; Mehmood, A.; Kim, K.-H. Metal-organic frameworks as an emerging tool for sensing various targets in aqueous and biological media. *TrAC Trends in Analytical Chemistry* **2019**, 120, 115654.
17. Munir, S.; Sattar, H.; Nadeem, A.; **Azam, M.** Thermal and kinetic performance analysis of corncobs, Falsa sticks, and Chamalang coal under oxidizing and inert atmospheres. *Energy Sources, Part A: Recovery, Utilization, and Environmental Effects* **2017**, 39, 775-782.
18. Wasim, M.; Sabir, A.; Shafiq, M.; Islam, A.; **Azam, M.**; Jamil, T. Mixed matrix membranes: Two step process modified with electrospun (carboxy methylcellulose sodium salt/sepiolite) fibers for nanofiltration. *Journal of Industrial and Engineering Chemistry* **2017**, 50, 172-182
19. Setoodeh Jahromy, S.; Jordan, C.; **Azam, M.**; Werner, A.; Harasek, M.; Winter, F. Fly ash from municipal solid waste incineration as a potential thermochemical energy storage material. *Energy & Fuels* **2019**, 33, 5810-5819
20. Setoodeh Jahromy, S.; Jordan, C.; **Azam, M.**; Werner, A.; Harasek, M.; Winter, F. Fly ash from municipal solid waste incineration as a potential thermochemical energy storage material. *Energy & Fuels* **2019**, 33, 5810-5819

21. Munawar, M.A.; Schubert, D.W.; Khan, S.M.; Rehman, M.A.U.; Gull, N.; Islam, A.; Sabir, A.; Shafiq, M.; Haider, B.; **Azam, M.** Investigation of functional, physical, mechanical and thermal properties of TiO<sub>2</sub> embedded polyester hybrid composites: A design of experiment (DoE) study. *Progress in Natural Science: Materials International* **2018**, *28*, 266-274.
22. Setoodeh Jahromy, S.; **Azam, M.**; Huber, F.; Jordan, C.; Wesenauer, F.; Huber, C.; Naghdi, S.; Schwendtner, K.; Neuwirth, E.; Laminger, T. Comparing Fly Ash Samples from Different Types of Incinerators for Their Potential as Storage Materials for Thermochemical Energy and CO<sub>2</sub>. *Materials* **2019**, *12*, 3358.
23. Mahmood, A.; Hussain Syed, J.; Raza, W.; Tabinda, A.B.; Mehmood, A.; Li, J.; Zhang, G.; **Azam, M.** Human Health Risk Assessment by Dietary Intake and Spatial Distribution Pattern of Polybrominated Diphenyl Ethers and Dechloran Plus from Selected Cities of Pakistan. *International journal of environmental research and public health* **2020**, *17*, 9543.
24. Farooq, A.; **Azam, M.**; Deen, K. Corrosion behavior of stainless steel weldments in physiological solutions. *Materials Research Express* **2018**, *5*, 015401.
25. Wesenauer, F.; Jordan, C.; Pichler, M.; Frei, A.; **Azam, M.**; Setoodeh Jahromy, S.; Harasek, M.; Winter, F. An Unreacted Shrinking Core Model Serves for Predicting Combustion Rates of Organic Additives in Clay Bricks. *Energy & Fuels* **2020**, *34*, 16679-16692.
26. Rashid, M. U., Daud, W. M. A. W., Mohidem, N. A., Mohamad, M. B., Akhtar, J., **Azam, M.**, & Miran, W. (2022). Methane dry reforming with CO<sub>2</sub> over ceria supported Ni catalyst prepared by reverse microemulsion synthesis. *Fuel*, *317*, 123433.
27. Tahir, K.; Miran, W.; Nawaz, M.; Jang, J.; Shahzad, A.; Moztahida, M.; Kim, B.; **Azam, M.**; Jeong, S.E.; Jeon, C.O. Investigating the role of anodic potential in the biodegradation of carbamazepine in bioelectrochemical systems. *Science of the total environment* **2019**, *688*, 56-64.
28. Khan, M. S., Raza, W., Gul, H., Hussain, M., Malik, B., **Azam, M.**, & Winter, F. (2021). A study on the reaction kinetics of anaerobic microbes using batch anaerobic sludge technique for beverage industrial wastewater. *Separations*, *8*(4), 43.
29. Khan, M. A., Raza, N., Manzoor, S., Shuja, R., Raza, H., Khan, M. I., **Azam, M.**, & Shanableh, A. (2023). Experimental design by response surface methodology for efficient cefixime uptake from hospital effluents using anion exchange membrane. *Chemosphere*, *311*, 137103.
30. Dai, Y., Gul, H., Sun, C., Tan, L., Guo, Y., Raza, W., Hussain, A., Pan, J., **Azam, M.**, & Zhu, W. (2023). Ultrafast synthesizing nanoflower-like composites of metal carbides and metal oxyhydroxides towards high-performance supercapacitors. *Electrochimica Acta*, *438*, 141575.
31. Khan, N. A., Bokhari, A., Mubashir, M., Klemeš, J. J., El Morabet, R., Khan, R. A., Alsubih, M., **Azam, M.**, Saqib, S., & Mukhtar, A. (2022). Treatment of Hospital wastewater with submerged aerobic fixed film reactor coupled with tube-settler. *Chemosphere*, *286*, 131838.
32. Ahmad, A., Hussain, F., Deen, K., Ahmad, R., Ali, L., Kamran, M., & **Azam, M.** (2014). Corrosion behavior of X-70 pipe steel in crude oil environments depending upon surface characteristics. *Arabian Journal for Science and Engineering*, *39*, 5393-5404.
33. Manzoor, S., Yasmin, G., Raza, N., Fernandez, J., Atiq, R., Chohan, S., Iqbal, A., Manzoor, S., Malik, B., & Winter, F. (2021). Synthesis of polyaniline coated magnesium and cobalt oxide nanoparticles through eco-friendly approach and their application as antifungal agents. *Polymers*, *13*(16), 2669.
34. Ahmad, M., Althobaiti, A. T., Ali, W., Masood, K., Ramadan, M. F., Chaudhary, B.,

- Zafar, M., Akhtar, M. S., Sultana, S., & Zahmatkesh, S. (2023). Membrane-processed honey samples for pollen characterization with health benefits. *Chemosphere*, 319, 137994.
35. A. Shafeeq, A Muhammad, R.U.Khan, **M. Azam** . Application of Heuristics to solve scheduling problems of intermediate transfer policies in multiproduct chemical batch “Journal of Quality and Technology Management”
  36. Aamir Ijaz, Shahid Munir, Raffi Ullah Khan, **Mudassar Azam**, Anum Nawaz, Ahsan Saleem .Development of technologies for removal of Arsenic from potable drinking water“Journal of the Pakistan Institute of Chemical Engineers”
- Further publications can be reached from google scholar link  
<https://scholar.google.com/citations?hl=en&user=Gbu7mn8AAAAJ>

### Poster and Conference Participation

1. **Mudassar Azam\***, Saman Setoodeh Jahromy, Franz Winter  
***Municipal Solid Waste Management in Lahore, Pakistan: Characterization & Energy Content*** Recy&DepoTech 2018, 7.-9. November 2018, Leoben, Austria.
2. **Mudassar Azam**, Franz Winter  
***Waste Management system in Pakistan; Country report***  
 07-IRRC Waste To Energy Conference, 2017, Vienna, Austria.
3. Saman Setoodeh Jahromy\*, Christian Jordan, **Mudassar Azam**, Andreas Werner, Michael Harasek, Franz Winter.  
***Fly Ash as Thermochemical and CO<sub>2</sub> Storage Material***  
 Österreichisches IEA-Wirbelschichttreffen, 3.-5. April 2019, Bruck an der Mur, Austria
4. Saman Setoodeh Jahromy\*, Christian Jordan, **Mudassar Azam**, Andreas Werner, Michael Harasek, Franz Winter  
***Fly Ash from Municipal Solid Waste Incineration as a Potential Thermochemical Energy Storage Material***  
 The 27th International Conference on the Impact of Fuel Quality on Power Production and the Environment, September 24th – 28th, 2018 Lake Louise, AB, Canada.
5. **Mudassar Azam**, Iffat Noreen, Hamda Ayub & Muhammad Furqan Mughal  
***Overview of the waste management system and its perspective in major cities of Pakistan*** IRRC Waste TO Energy Conference, 5.-6. September, 2022 in Vienna, Austria
6. **Mudassar Azam** “*Waste to Applications; circular Economy, Sustainability Conference 2024, at GIKI, Lahore Pakistan*”

### Achievement and Awards

- IRCA Qualified lead auditor ISO 9001:2008
- Won scholarship by Rotary International for group study exchange for Indiana, USA
- Got Merit Scholarship in 1<sup>ST</sup>, 2<sup>ND</sup> & 3<sup>RD</sup> Professional from University of the Punjab
- 2<sup>nd</sup> Position in M.Sc. Chemical Engineering, University of the Punjab
- Attended International Solid Wastes Association (ISWA) study tour **Waste-to-Energy**, Austria and Germany
- Best PhD paper award

## **Professional Membership**

### **Members:**

- Pakistan Engineering Council (Registered Engineer CHEM/7380)
- Pakistan Institute of Chemical Engineers
- Member of International Solid Wastes Association (ISWA), Austria

## **Courses and Participation**

- Professional Competency Enhancement Program for Teachers (PCEPT) organized by HEC
- Participated in Waste to Energy study tour, organized by ISWA, Austria and Germany
- Participated in Symposium on Engineering Sciences, University of the Punjab